

SPECIAL SECTION PAPER

Extremely Robust Rollable AMOLED Display With Very Small Waviness of Surface Using Smart Module and Panel Structure

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ABSTRACT

An extremely robust rollable AMOLED display with excellent surface quality (very low waviness) using smart module and panel structure was proposed. Using high-energy damping OCA (Optical Clear Adhesive) of PUA (polyurethane acrylate) elastomer with high loss tangent instead of conventional PSA of silicone significantly increased pen drop height from 4 to 7 cm and angle of pendulum in Charpy test from 68° to 120° (maximum angle of Charpy test equipment used in this study) when estimating side-impact property of OCA of PUA elastomer with high loss tangent. Applying hybrid-polymer OC (overcoating) material directly on and under panel remarkably improved ball drop height from 5 to 21 cm. Replacement of symmetric multi-joint bar of stainless steel with asymmetric multijoint bar of CFRP (carbon fiber reinforced plastics) greatly enhanced surface quality of rollable display module and lessened K_c (one of evaluation standard for waviness) from 0.80 to 0.44 after 240 h at the condition of 60°C/93% RH, where smaller K_c means smaller waviness of rollable display module. We also demonstrated IT and automotive rollable AMOLED rollable display with rolling radius of 7 and 50 mm, respectively. Touch integrated panel was also applied for the rollable display.

1 | Introduction

Nowadays, rollable AMOLED displays have been receiving a lot of attention as a candidate of promising next-generation displays in near future thanks to its superb portability based on its extremely expandable property from compact storage space. Rollable AMOLED display requires several crucial characteristics such as high flexibility for small rolling radius, excellent robustness of panel and stacked module about external impact from ball drop and pen drop, outstanding recoverability or property of no creep after a period of rolling stress and rolled stress, and high surface quality of low waviness of rollable display module where waviness is usually caused by low flexibility and poor recoverability and creep problem. To achieve these key technologies, very flexible, recoverable, and high energy damping

materials must be developed, and a smart stacked module structure for the rollable display also has to be suggested.

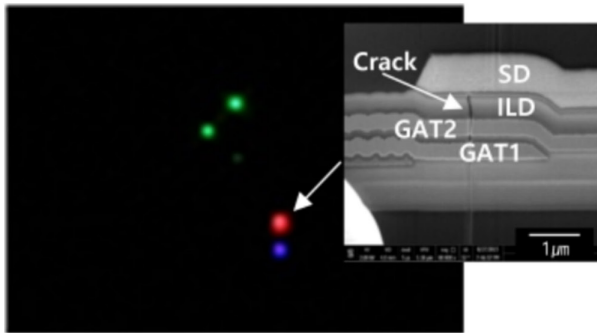
Besides these characteristics, rollable AMOLED displays have to possess good reliability under diverse environmental and mechanical test conditions. Most display companies have been developing rollable AMOLED displays in order to sustain these essential properties in various environmental test conditions such as high temperature, low temperature, and high temperature and high humidity and different mechanical test conditions such as cyclic rolling, static rolling, and a combination of cyclic rolling and static rolling [1–17]. We have worked on the robustness and surface quality of rollable AMOLED display and suggested a new smart module and panel structure, and a novel multijoint support with a symmetric CFRP (carbon fiber reinforced plastics) structure

in this study in order to considerably enhance surface quality (namely, reduce waviness) and increase several robustness properties including pen drop, ball drop, and side-impact property.

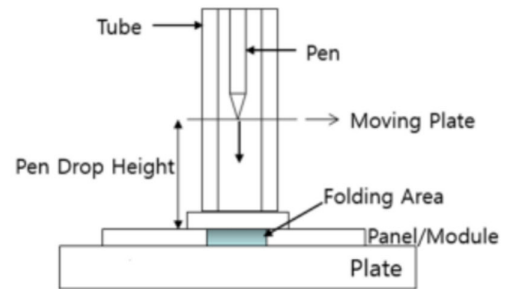
2 | Experiments and Results

Rollable AMOLED display usually tends to show very vulnerable characteristics against external impact since it does not use very thick glass window and instead is likely to adopt very thin glass window or plastic film window to perform good rollability. For this reason, rollable AMOLED display must improve robustness properties such as pen drop and ball drop in Figure 1. In case of ball and pen drop test, panel shows crack phenomenon when strain of panel goes beyond critical strain of panel where crack is initiated. Typically, initial crack starts from continuous inorganic layer or metal layer since these materials are very brittle (or have high modulus) and there is no area (space or structure) for these continuous layers to disperse or mitigate external impact energy. These cracks lead to dark pixel spot or bright pixel spot during pen drop and ball drop test as shown in Figure 1.

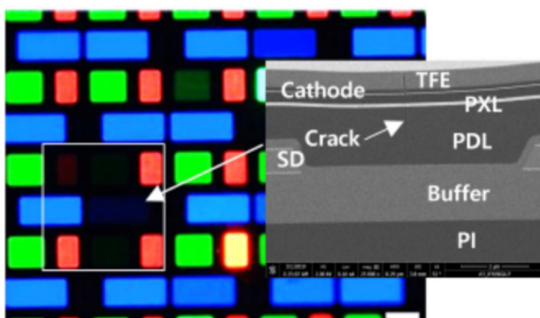
Therefore, it is definitely required that rollable AMOLED display must have a very robust module and panel structure to demonstrate excellent characteristics of pen drop and ball drop while it still has a thin module structure to hold rollability. In order to raise pen drop height and ball drop height of rollable AMOLED display, high impact energy damping OCA (Optical Clear Adhesive) materials consisting of PUA (polyurethane acrylate) elastomer with different loss tangent ($\tan\delta$) values were applied on polarizer in Figure 2. OCA material consisting of PUA elastomer with a higher loss tangent ($\tan\delta$) of 0.9 significantly improved pen drop height instead of using conventional PSA (pressure sensitive adhesive) material with a loss tangent ($\tan\delta$) of 0.3 since OCA material consisting of PUA elastomer with a higher loss tangent ($\tan\delta$) has a better external impact energy damping effect in collision. When we manufacture MES (multijoint embedded support) structure, multijoint bars made up of stainless steel (SUS) are embedded inside resin in Figure 2 and then resin of PUA elastomer is cured. AG, AR, and AF mean antiglare coating, antireflection coating, and antifingerprint coating, respectively. These diverse coating materials are used on the top of the window layer especially for automotive rollable display.



[Panel Damage in Pen Drop Test]

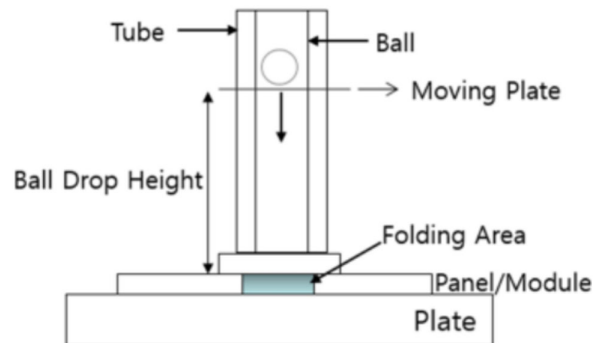


[Pen Drop Test Equipment]



[Panel Damage in Ball Drop Test]

(a)



[Ball Drop Test Equipment]

(b)

FIGURE 1 | (a) Pen drop test equipment and panel damage in pen drop test and (b) ball drop test equipment and panel damage in ball drop test for rollable AMOLED module structure.

Module Structure	[Reference] AGARAF Film Window PSA (0.04MPa) POL PSA Panel PSA Film PSA MES Metal Resin	AGARAF Film Window OCA (1.6MPa) POL PSA Panel PSA Film PSA MES Metal Resin
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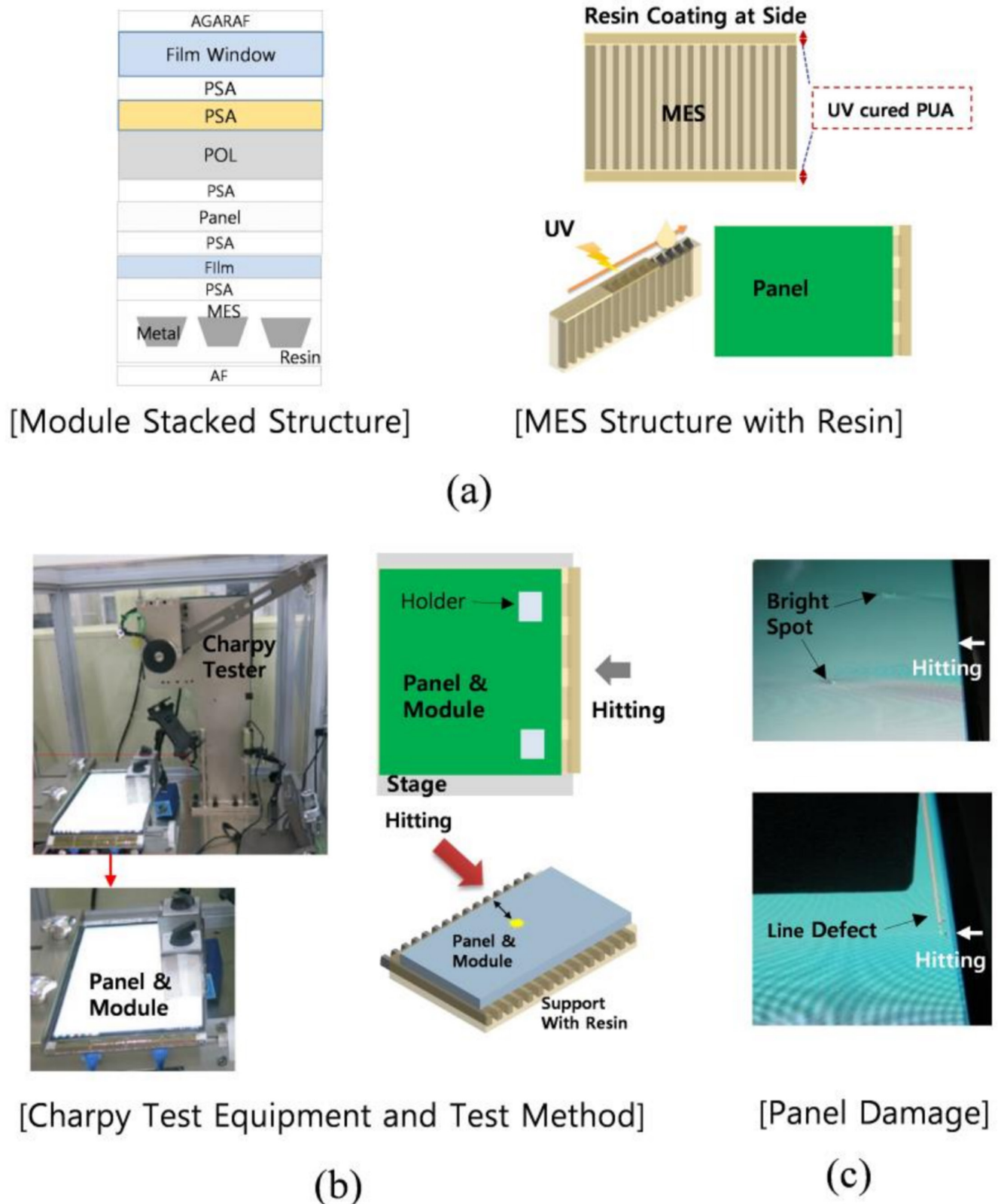


FIGURE 3 | (a) Module stacked structure and MES (multijoint embedded support) structure for Charpy test, (b) Charpy test equipment and test method, and (c) panel damage after Charpy test.

PSA of silicone with OCA material of PUA elastomer to display high impact energy damping property. The angle of pendulum of Charpy test equipment is defined as a maximum degree where no panel damage occurs. In case of high impact energy damping

OCA material of PUA elastomer with loss tangent ($\tan \delta$) of 0.90, there is no panel damage of line defect or bright spot even after thirty hitting in a row at angle of 120° of pendulum in Charpy test in Figure 4b. Impact energy was calculated using Equation (1)

where weight of pendulum (m) is 202g and arm length (L) is 30cm. When high impact energy damping OCA material of PUA elastomer with loss tangent ($\tan \delta$) of 0.90 was applied instead of conventional PSA of silicone, impact energy was raised up from 3788 gf · cm to 9091 gf · cm in Table 1.

Small deflection to sustain support property and surface quality after rolling are essential characteristics for rollable AMOLED display when manufacturing rollable display. MES is proposed to sustain rigid form and be not easily deflected when rollable display rolls out and be also easy to roll. MES must be light in weight as well for mobile application. Therefore, MES made up of CFRP is preferred due to less weight compared with SUS 304 (a kind of stainless steel) since density of CFRP is $\sim 1.8\text{g/cm}^3$ and density of SUS 304 is $\sim 7.7\text{g/cm}^3$ in Table 2. Because multijoint bar used

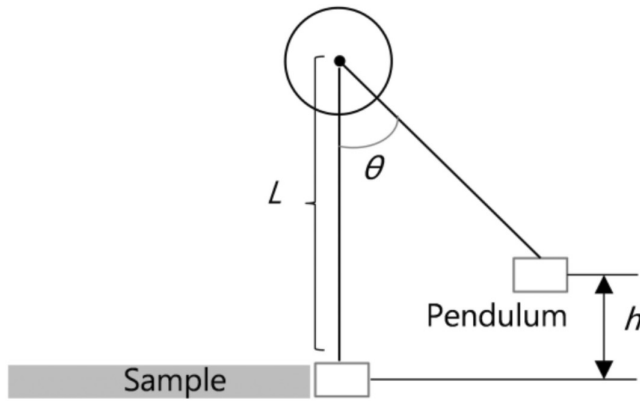


FIGURE 4 | Schematic diagram for Charpy test equipment.

TABLE 1 | Result for Charpy test using PSA of silicone and OCA of PUA elastomer.

Resin	Angle (°) for panel damage	Impact energy (gf · cm)
Silicone	68	3788
PUA	120	9091

TABLE 2 | Physical property of SUS 304 (stainless steel) and CFRP (carbon fiber reinforced plastics).

	SUS304	CFRP
Density (g/cm ³)	~ 7.7	~ 1.8
Flexural strength (MPa)	~ 900	~ 700
Tensile strength (MPa)	1020	2546
Modulus (GPa)	193	120

in this study has rectangular cross-section, extent of deflection height based on gravitational force coming from weight of multijoint bar itself can be compared for stainless steel and CFRP using Equation (2), and schematic diagram to calculate deflection height of multijoint bar rectangular cross-section is shown in Figure 5 [18]. Compared to weight of SUS 304, less weight of CFRP leads to lower deflection height because of smaller gravitational force coming from less weight when applying $F = mg$ in Equation (2), m is weight of multijoint bar when assuming the same dimension of multijoint bar of SUS 304 and CFRP and considering the density and modulus of SUS 304 and CFRP.

With respect to estimation of surface quality (or waviness), we used PMD (Phase Measuring Deflectometry) analysis method. Projected sinusoidal beam is illuminated to surface and reflected beam is detected on camera in Figure 6a [19]. Height and slope of surface are evaluated in Figure 6a. Different kinds of K values are able to be filtered based on different range of wavelength (or waviness width) in Figure 6b and then K (curvature) in each range of wavelength (or waviness width) is measured. The measured value of K (curvature) amounts to standard deviation of curvature and is the derivative of the slope in Figure 6c. Small K (standard deviation of curvature) means better surface quality, meaning less waviness. Because K_c is the most sensitive to human eyes at typical viewing distance of mobile and IT product of 40cm, K_c is widely used to evaluate surface quality of display (or waviness) as shown in Figure 6c [20, 21]. Replacing SUS 304 with CFRP reduced weight of module and panel and enhanced surface quality, lessening K_c value from 0.8 to 0.63 after 240h at the condition of 60°C/93% RH, which corresponds to less waviness, namely, better surface quality in Figure 7. It is assumed that less deflection of CFRP contributes to better surface quality.

Large MES could reduce strain of panel more than small multijoint bar where A of thickness is larger than C and B of width is larger than D. But there is a trade-off that while large multijoint bar can improve impact resistance property, it might also cause problem in achieving reliable rolling property at small rolling radius due to thick stacked module structure. Simulation result indicated that strain of panel at edge of multijoint bar embedded in resin is larger than strain of panel at center of multijoint bar embedded in resin with respect to external impact test in Figure 8.

Symmetric structure of CFRP of multijoint bar increased waviness on the surface of the rollable display during rolling in Figure 9. Large deformation of resin between symmetric multijoint bars happened and then triggered large waviness (or wrinkle deformation) on the surface of the rollable display module as shown in the cross-sectional X-ray image and optical surface image in Figure 9. However, asymmetric structure of CFRP of multijoint bar greatly decreased deformation of resin between asymmetric multijoint bars and therefore showed very good

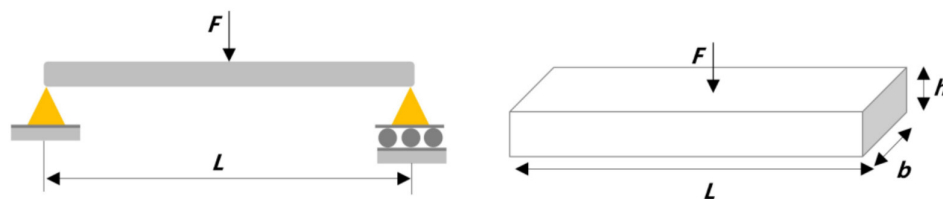


FIGURE 5 | Schematic diagram to calculate deflection of multijoint bar with rectangular cross-section.

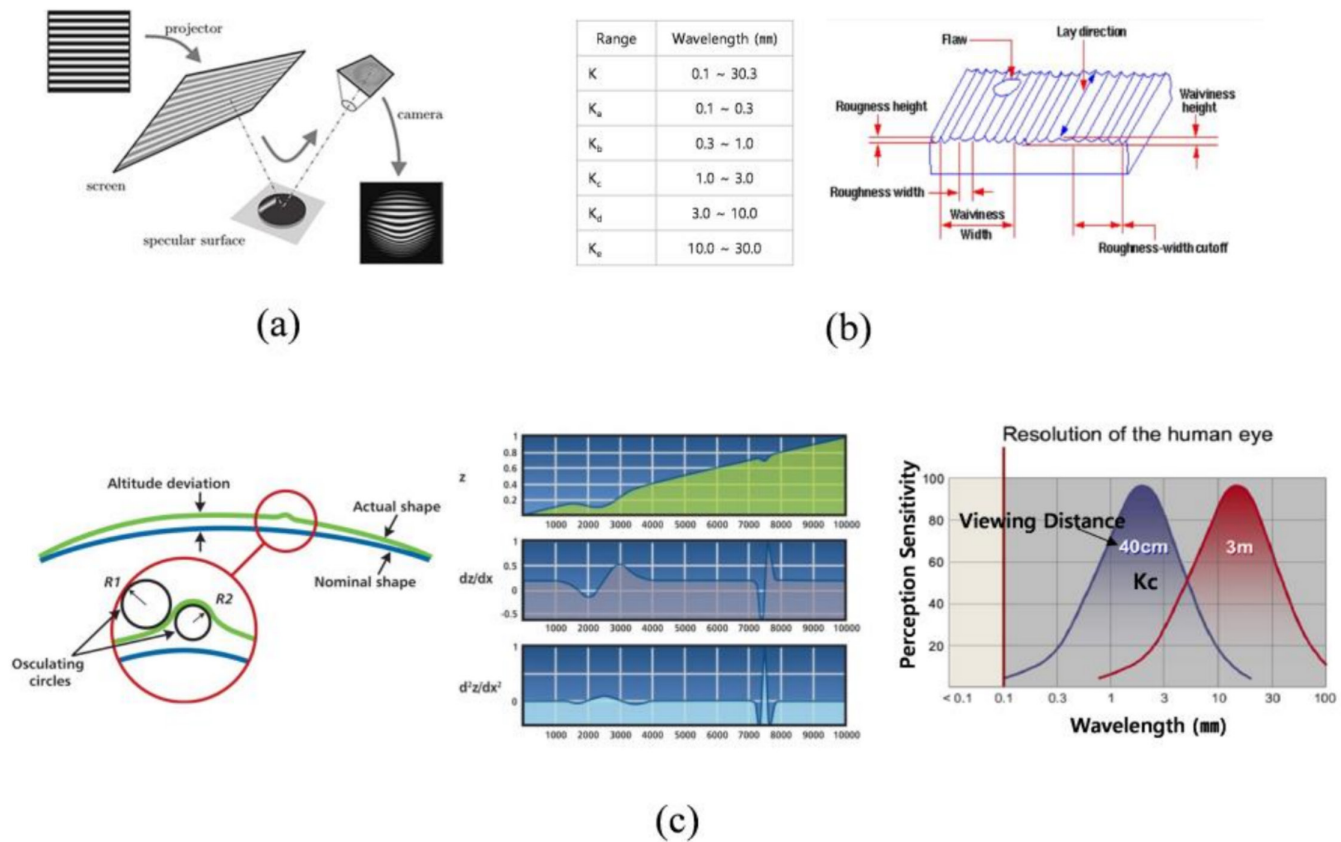


FIGURE 6 | (a) PMD (phase measuring deflectometry) method, (b) categories of K values based on different wavelength of corresponding waviness, and (c) evaluated curvature to be equal to derivative of slope, and perception sensitivity of K values in different range of wavelength.

Module Structure	[SUS]	[CFRP]
	AGARAF Film Window PSA POL PSA Panel PSA Film PSA MES SUS Resin AF	AGARAF Film Window PSA POL PSA Panel PSA Film PSA MES CFRP Resin AF
K _c (60°C/93%RH, 240Hr @ R30)	0.8	0.63
Weight (g) (12.4")	296	122

FIGURE 7 | Result for surface quality of waviness (K_c) after 60°C/93% RH for 240 h at rolling radius of 30 mm for MES of SUS 304 and CFRP respectively.

surface quality of the rollable display module as shown in the cross-sectional X-ray image and optical surface image in Figure 9 due to reduced strain of resin coming from wider gap between the bottom area of the asymmetric structure of CFRP of the multijoint bar in the area of inner circle of MES in Figure 9.

In conclusion, the replacement of the symmetric multijoint bar of CFRP with the asymmetric multijoint bar of CFRP greatly enhanced the surface quality of the rollable display module and further lessened Kc (the first derivative of slope or waviness) from 0.50 to 0.44 after 240 h at the condition of 60°C/93% RH.

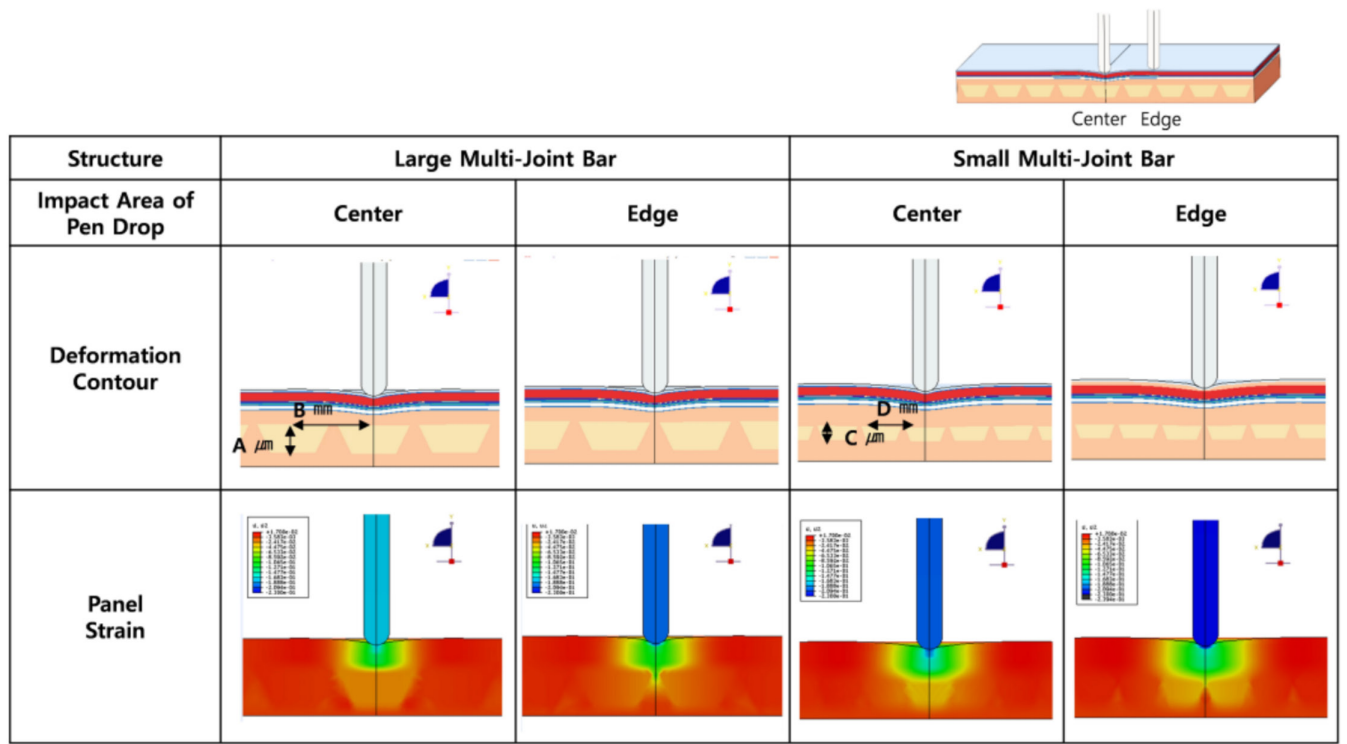


FIGURE 8 | Simulation result of strain in ball drop test for large and small size of multijoint bar where A is larger than C and B is larger than D.

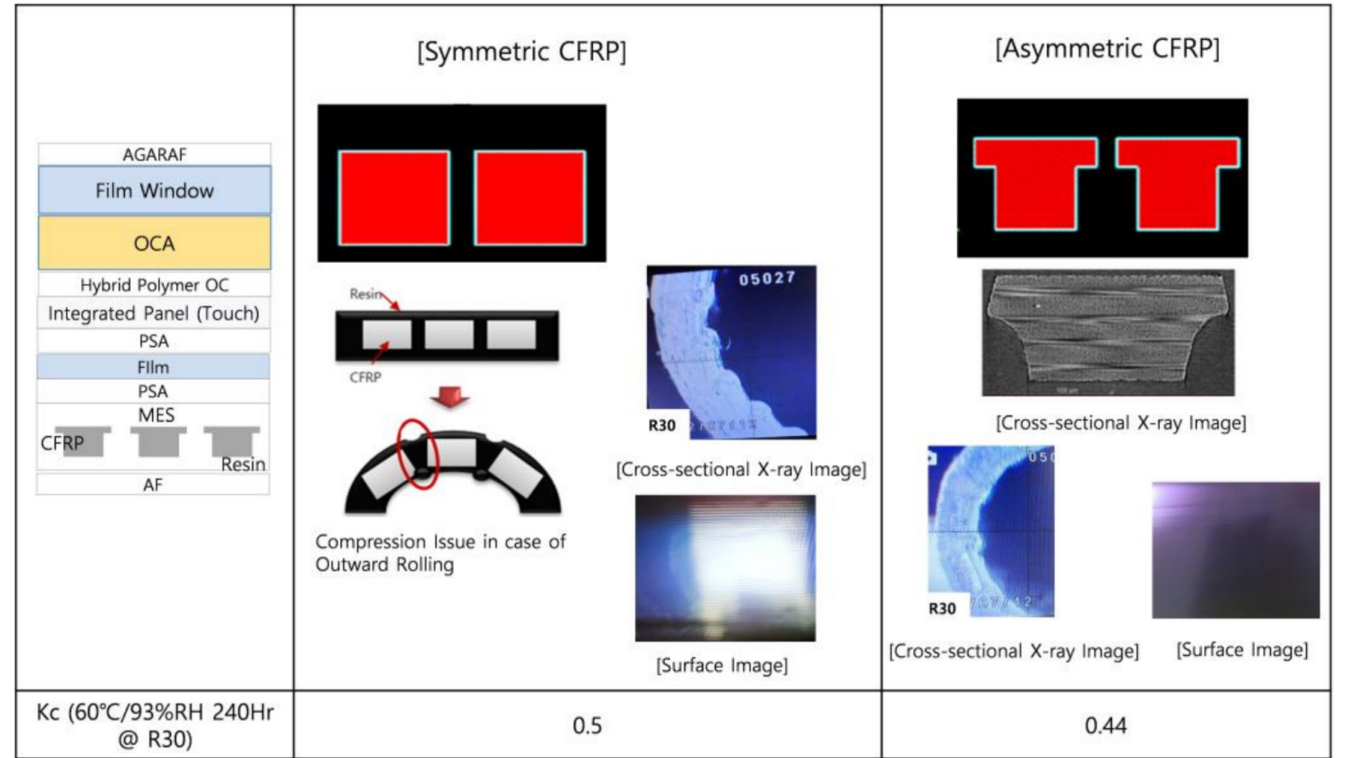


FIGURE 9 | Waviness result for symmetric and asymmetric CFRP (carbon fiber reinforced plastics) multijoint bar after 60°C/93% RH for 240 h at rolling radius of 30 mm.

In order to improve ball drop height, new hybrid-polymer OC (overcoating) material of thickness of 30μm along with high impact energy damping OCA of PUA elastomer was

TABLE 3 | Modulus of hybrid polymer and polymer of over coating.

Material of over coating	Modulus (GPa)
Hybrid polymer	4.4
Polymer	1.3

introduced on panel and under panel. This hybrid-polymer OC material (modulus; 4.4GPa) tremendously increased ball drop height from 5 to 21cm compared with conventional polymer OC material (modulus; 1.3GPa) in Table 3. High-energy damping OCA of PUA elastomer is located on hybrid-polymer or conventional polymer OC material. High impact energy damping OCA of PUA elastomer can absorbed impact energy and lessen impact energy in collision with ball. The hybrid- polymer OC material is able to lessen deflection height and reduce strain of panel in Figure 10 [8, 9]. In this study,

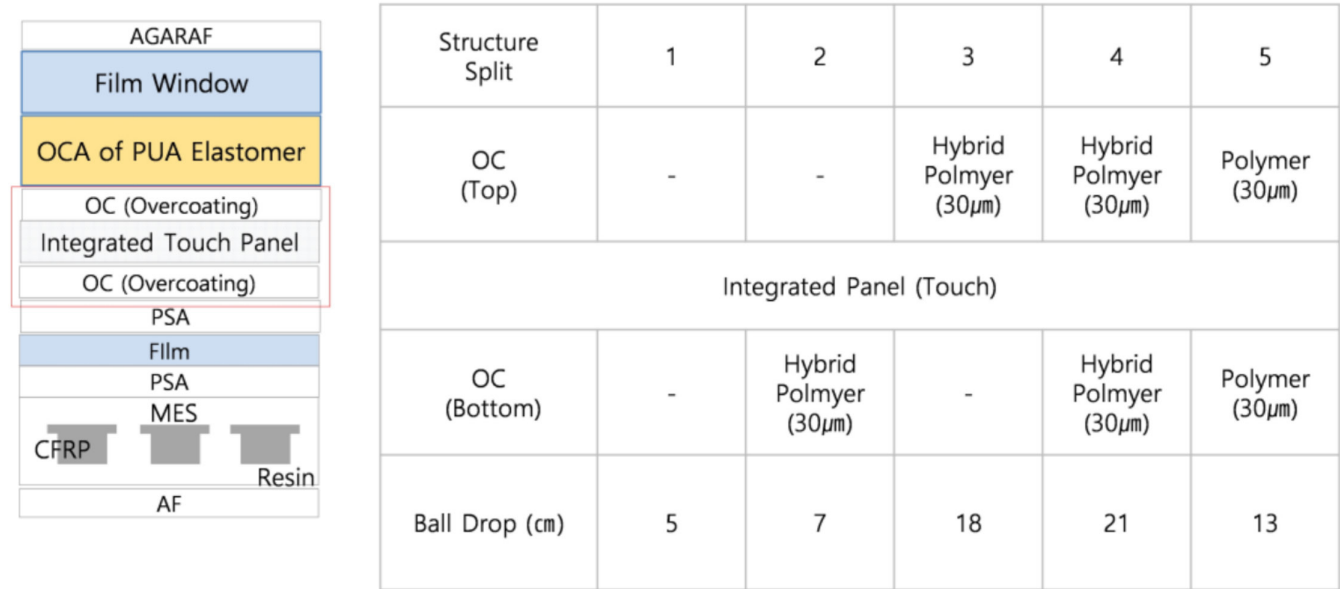


FIGURE 10 | Ball drop test results for different integrated touch panel structures using hybrid-polymer OC (over coating) material and polymer OC (over coating) material.

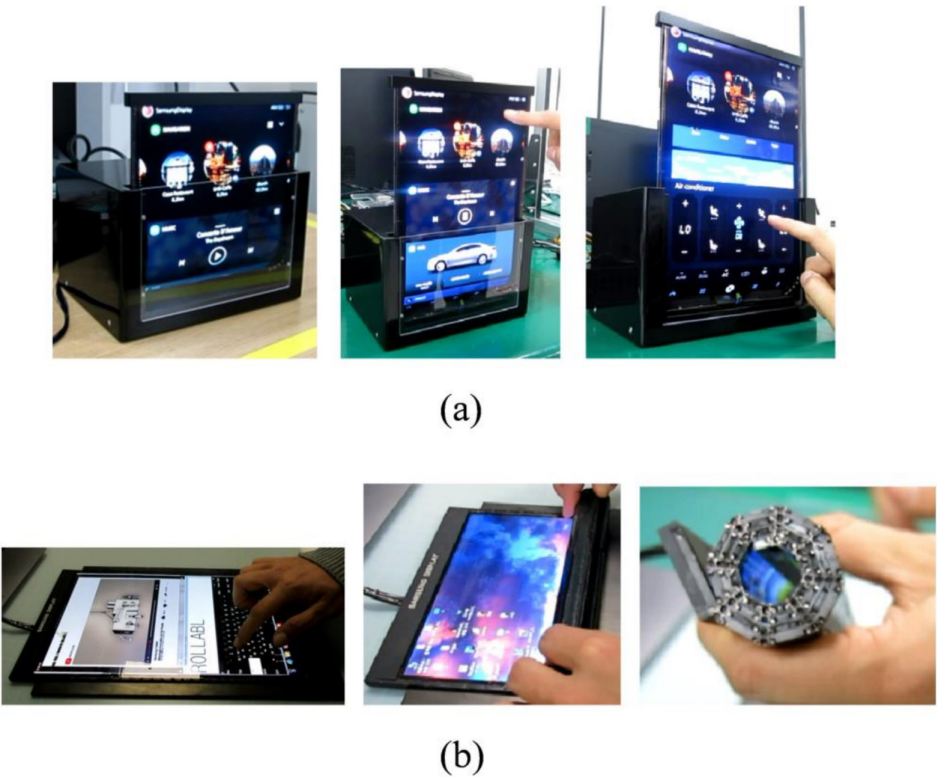


FIGURE 11 | 12.4-in. rollable AMOLED display for (a) automotive and (b) IT applications.

asymmetric structure of CFRP consisting of MES was also applied along with this new smart touch integrated panel and stacked module structure.

This extremely robust proto-type 12.4-in. touch integrated rollable AMOLED display with outstanding surface quality using smart module and panel structure was demonstrated for automotive and IT applications. Automotive rollable display can afford three kinds of panel size on the extent of panel expansion and IT rollable display can propose possibility of excellent portability (Figure 11).

3 | Conclusion

Remarkable robust rollable AMOLED display with excellent surface quality was suggested by introducing high energy damping OCA (Optical Clear Resin) material of PUA (Polyurethane Acrylate) elastomer, hybrid-polymer OC (overcoating) material, and asymmetric CFRP multibar structure. This noble panel and module structure for rollable display significantly improved pen drop height from 3 to 7 cm, ball drop height from 5 to 21 cm, and surface quality (Kc) from 0.8 to 0.44 after 240 h at the condition of 60°C/93% RH. We also demonstrated the extremely robust proto-type 12.4-in. touch integrated rollable AMOLED display for IT and automotive applications with rolling radius of 7 and 50 mm, respectively.

$$E = \frac{1}{2} mV^2 = \frac{1}{2} mgh = mgL (1 - \cos\theta) \quad (1)$$

$$\delta_c = \frac{FL^3}{48EI} \quad I = \frac{bh^3}{12} \quad (2)$$

where δ_c is deflection height, F is force acting on the center of the multijoint bar, L is length of the beam between the multijoint bar, E is modulus of elasticity of the multijoint bar, I is area moment of inertia of cross-section of the multijoint bar, b is width of the multijoint bar, and h is thickness of the multijoint bar.

Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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